Hints: Trigonometry – 3. The substitution $x = 2\cos\theta$ and Chebyshev polynomials

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The recurrence $T_{n+1} = 2xT_n - T_{n-1}$ comes from $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$; induction gives degree n and leading coefficient 2^{n-1} ($n \geqslant 1$).

Hint 2. $2^{1-n}T_n$ is monic with maximum modulus 2^{1-n}, attained with alternating signs at the $n+1$ points $x_k = \cos\frac{k\pi}{n}$, $k = 0, \dots, n$. Suppose a monic p of degree n had $\max_{[-1,1]}|p| < 2^{1-n}$ and look at $q = 2^{1-n}T_n - p$.

Hint 3. $\deg q \leqslant n-1$, and $q(x_k)$ has the sign of $(-1)^k$, so q has a zero in each of the n intervals (x_{k+1}, x_k): $q \equiv 0$, contradiction. For uniqueness (when $\max|p| = 2^{1-n}$) the inequalities become weak: count zeros with multiplicity, noting that a zero of q at an interior x_k is at least double because q does not change sign there. The last claim of the problem is then the theorem itself.

Problem 2

Hint 1. Differentiate $T_n(\cos\theta) = \cos n\theta$: $T_n'(\cos\theta) = \frac{n\sin n\theta}{\sin\theta} = nU_{n-1}(\cos\theta)$.

Hint 2. $|\sin n\theta| \leqslant n|\sin\theta|$ (induction, or the addition formula) gives $|T_n'| \leqslant n^2$ on $[-1,1]$, with equality at $\theta \to 0, \pi$, i.e. $x = \pm 1$; and $|\sin n\theta| \leqslant 1$ gives $|T_n'(x)| \leqslant n/\sqrt{1-x^2}$.

Hint 3. For the trigonometric Bernstein inequality reduce to real T (rotate by $e^{i\gamma}$ so that $T'(\theta_0)$ is real and take the real part). Assume $\|T\|_\infty < 1$ but $T'(\theta_0) \geqslant n$, and compare T with the extremal "Chebyshev" function $S(\theta) = \sin n(\theta - \alpha) = T_n\big(\cos(\theta - \alpha - \frac{\pi}{2n})\big)$, with α chosen so that $S(\theta_0) = T(\theta_0)$ on an increasing branch of S.

Hint 4. $D = S - T$ is a trigonometric polynomial of degree n. At the $2n$ extrema of S (values ± 1) D alternates in sign, giving $2n$ zeros; in the increasing branch through θ_0 one has $D(\theta_0) = 0$ and $D'(\theta_0) = S'(\theta_0) - T'(\theta_0) \leqslant n - n = 0$ although D goes from negative to positive, giving at least 2 extra zeros. More than $2n$ zeros forces $D \equiv 0$, contradicting $\|T\| < 1$.

Problem 3

Note. Since $a_1 = a$, the literal condition "$a_n \in \mathbb{Z}$ for all n" just means $a \in \mathbb{Z}$, and then the claim is easy. The hints treat this and also the more interesting reading "$a_n \in \mathbb{Z}$ for all large n".

Hint 1. Substitute: if $|a| \leqslant 2$ write $a = 2\cos\theta$, if $|a| > 2$ write $a = t + t^{-1}$ with real t, $|t| > 1$. Then $a_n = 2\cos(2^{n-1}\theta)$, resp. $a_n = t^{2^{n-1}} + t^{-2^{n-1}}$.

Hint 2. Behaviour: for $|a| > 2$, $a_n \to +\infty$ (doubly exponentially). For $|a| \leqslant 2$ the sequence is the doubling map $\theta \mapsto 2\theta$ on the circle: eventually periodic iff $\theta/\pi \in \mathbb{Q}$, constant 2 eventually iff θ/π is a dyadic rational, and otherwise bounded without being eventually periodic (for θ/π irrational it is dense in $[-2,2]$ for almost every θ, but not always).

Hint 3. Integrality: if $a \in \mathbb{Z}$, then either $|a| \leqslant 2$, i.e. $a \in \{0, \pm 1, \pm 2\}$, or t is a root of $x^2 - ax + 1$, a monic integer polynomial with constant term 1, so t is a (quadratic) unit.

Hint 4. If only $a_N \in \mathbb{Z}$ (hence $a_n \in \mathbb{Z}$ for $n \geqslant N$), then $u = t^{2^{N-1}}$ satisfies $u^2 - a_N u + 1 = 0$, so t is a root of $x^{2^N} - a_N x^{2^{N-1}} + 1$ and is again an algebraic unit (for $|a| \leqslant 2$ take $t = e^{i\theta}$).

Problem 4

Hint 1. Use the closed forms (with $w = \sqrt{x^2-1}$)

$$T_n(x) = \frac{(x+w)^n + (x-w)^n}{2}, \qquad U_{n-1}(x) = \frac{(x+w)^n - (x-w)^n}{2w}$$

(they satisfy the recurrence and the initial values).

Hint 2. At $x = \frac{i}{2}$: $\sqrt{x^2-1} = \frac{i\sqrt{5}}{2}$ and $x \pm \sqrt{x^2-1} = i\varphi,\ i\psi$ with $\varphi, \psi = \frac{1\pm\sqrt{5}}{2}$. Binet's formulas $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$, $L_n = \varphi^n + \psi^n$ finish the computation.

Hint 3. Catalan's identity $F_n^2 - F_{n-r}F_{n+r} = (-1)^{n-r}F_r^2$ is the image of the polynomial identity $U_{n-1}^2 - U_{n-r-1}U_{n+r-1} = U_{r-1}^2$, which is $\sin^2 n\theta - \sin(n-r)\theta\sin(n+r)\theta = \sin^2 r\theta$ divided by $\sin^2\theta$. Evaluate at $x = \frac{i}{2}$ and track the powers of i. (The analogous identity for T_n, $T_n^2 - T_{n-r}T_{n+r} = 1 - T_r^2$, gives the Lucas version.)

Problem 5

Hint 1. Put $x = \cosh\theta$. Then $T_k(x) = q_1$ and $T_{k+1}(x) = q_2$ are rational, so x is a common root of the rational polynomials $P(X) = T_k(X) - q_1$ and $Q(X) = T_{k+1}(X) - q_2$. Their gcd has rational coefficients (Euclid's algorithm), so it suffices to show that x is their only common complex root, and a simple one.

Hint 2. Find all complex roots: $T_k(\cosh w) = \cosh kw$ for complex w, so the roots of P are $\cosh(\theta + \frac{2\pi ij}{k})$, $j = 0, \ldots, k-1$, and these k numbers are distinct because $\theta > 0$. Similarly for Q with $k+1$.

Hint 3. $\cosh u = \cosh v$ iff $u \equiv \pm v \pmod{2\pi i}$. The minus sign would make 2θ purely imaginary; the plus sign gives $\frac{j}{k} - \frac{l}{k+1} \in \mathbb{Z}$, which forces $j = l = 0$ since $\gcd(k, k+1) = 1$. So $\gcd(P, Q) = X - x$, and $x \in \mathbb{Q}$.